

1.1. APACHE NIFI KASU PRAKTIKOAK (5-6-7)

Aurkibidea

Apache NiFi 5. kasua: CSV JSON bihurtu.....	2
Apache NiFi 6. kasua: MariaDB → MongoDB	13
6. kasuaren 2. aldaera: PutMongoRecord	21
Apache NiFi 7. kasua: AEMET Datu-lakua.....	25

Apache NiFi 5. kasua: CSV JSON bihurtu

Helburua: CSV fitxategiak JSON-era bihurtu ConvertRecord prozesadorearekin, dena **prozesu-talde batean** sartuta (sarrera eta irteera-atakekin).

Fluxua: GetFile → ConvertRecord → UpdateAttribute → PutFile

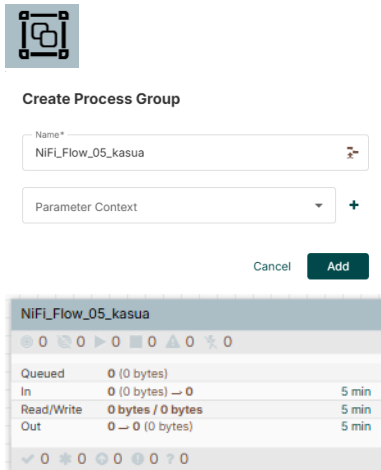
- **GetFile:** Sarrera-karpetatik CSV fitxategiak hartu
- **ConverRecord:** CSV → JSON bihurketa
- **UpdateAttribute:** filename atributua .csv → .json
- **PutFile:** Irteera-karpetan JSON fitxategia idatzi

Bihurketa: CSVReader (sarrera) → JsonRecordSetWriter (irteera)

Fitxategi-izena aldatu: .csv extensioa .json-ekin ordezkatu (UpdateAttribute)

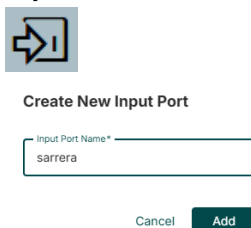
1. **Process Group** berria sortu:

ProcessGroup arrastatu, ariketa honi dagozkion elementuak bertan sartzeko. Dagokion izena jarri eta **Add** sakatu.



Sortuta dagoenean, klik bikoitza egin **Process Group**-aren gainean, bertan sartzeko.

2. **Input Port** sarrera-portua gehitu:



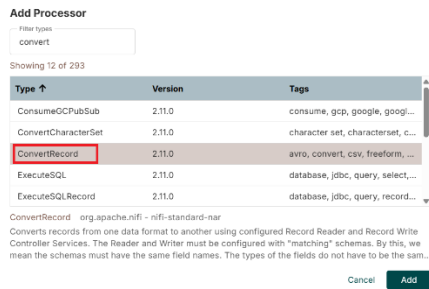
3. **ConvertRecord** prozesadorea:

Gehitu **ConvertRecord** prozesadorea, irteera-karpetan idazteko.

- a. **Processor** arrastatu.



b. **ConverRecord** aukeratu eta **Add** sakatu.

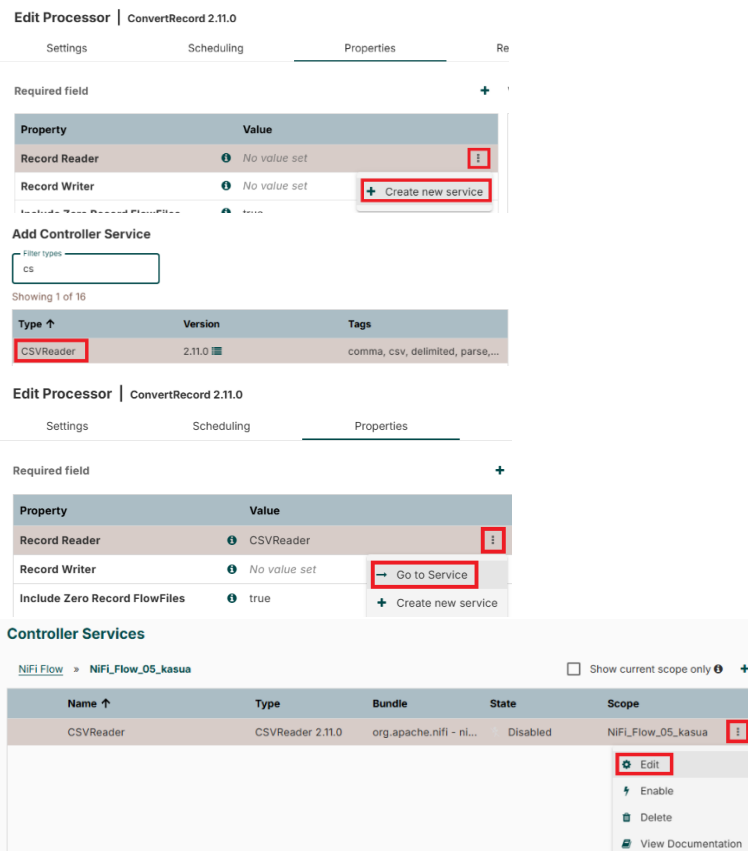


c. Klik bikoitza egin **ConvertRecord**-en gainean, propietateak aldatzeko.

	ConvertRecord ConvertRecord 2.11.0 org.apache.nifi - nifi-standard-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- **Record Reader: CSVReader**
 - **CSVReader (Controller Service):**
 - Schema Access Strategy: **Use String Fields From Header**
 - Value Separator: ;
 - Treat First Line as Header: **true**
- CSVReader Controller Service → **Enable**



Edit Controller Service

Settings Properties

Required field +

Property	Value
Schema Access Strategy	Use String Fields From Header
CSV Parser	Apache Commons CSV
Date Format	No value set
Time Format	No value set
Timestamp Format	No value set
CSV Format	Custom Format
Value Separator	;
Record Separator	\n
Treat First Line as Header	true
Ignore CSV Header Column Names	false

Controller Services

NIFI Flow > NIFI_Flow_05_kasua Show current scope only

Name	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-r...	Disabled	NIFI_Flow_05_kasua

Edit
Enable
Delete
View Documentation

- **Record Writer: JsonRecordSetWriter**
 - **JsonRecordSetWriter (Controller Service):**
 - Schema Access Strategy: Use String Fields From Header
 - Value Separator: ;
 - Treat First Line as Header: true

JsonRecordSetWriter Controller Service → **Enable**

Edit Processor | ConvertRecord 2.11.0

Settings Scheduling Properties R

Required field +

Property	Value
Record Reader	CSVReader
Record Writer	No value set
Include Zero Record FlowFiles	true

Create new service

Add Controller Service

Filter types
json

Showing 1 of 7

Type	Version	Tags
JsonRecordSetWriter	2.11.0	json, record, recordset, result...

Edit Processor | ConvertRecord 2.11.0

Settings Scheduling Properties F

Required field +

Property	Value
Record Reader	CSVReader
Record Writer	JsonRecordSetWriter
Include Zero Record FlowFiles	true

Go to Service
Create new service

Controller Services

NIFI Flow > NIFI_Flow_05_kasua Show current scope only

Name	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-r...	Enabled	NIFI_Flow_05_kasua
JsonRecordSetWriter	JsonRecordSetWriter ...	org.apache.nifi - nifi-r...	Disabled	NIFI_Flow_05_kasua

Edit
Enable
Delete
View Documentation

Edit Controller Service

Settings

Properties

Required field

+

Property	Value
Schema Access Strategy	Inherit Record Schema
Date Format	No value set
Time Format	No value set
Timestamp Format	No value set
Pretty Print JSON	false
Suppress Null Values	Never Suppress
Allow Scientific Notation	false
Output Grouping	Array
Compression Format	none
Serialized JSON Input Handling	Enabled

Controller Services

Nifi Flow > NIFLFlow_05_kasua

Show current scope only

+

Name	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-r...	Enabled	NIFLFlow_05_kasua
JsonRecordSetWriter	JsonRecordSetWriter ...	org.apache.nifi - nifi-r...	Disabled	NIFLFlow_05_kasua

Edit

Enable

Delete

View Documentation

Edit Processor | ConvertRecord 2.11.0

Settings

Scheduling

Properties

Required field

+

Property	Value
Record Reader	CSVReader
Record Writer	JsonRecordSetWriter
Include Zero Record FlowFiles	true

e. Relationships atalean:

- Automatically Terminate/Retry Relationships:
 - failure: terminate

Edit Processor | ConvertRecord 2.11.0

Settings

Scheduling

Properties

Relationships

Automatically Terminate/Retry Relationships

failure

terminate

retry

If a FlowFile cannot be transformed from the configured input format to the configured output format, the unchanged FlowFile will be routed to this relationship

success

terminate

retry

FlowFiles that are successfully transformed will be routed to this relationship

5

4. **UpdateAttribute** prozesadorea:
Gehitu **UpdateAttribute** prozesadorea.

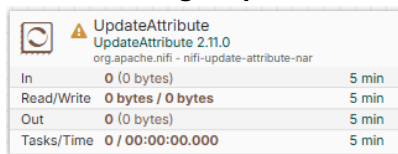
a. **Processor** arrastatu.



b. **UpdateAttribute** aukeratu eta **Add** sakatu.

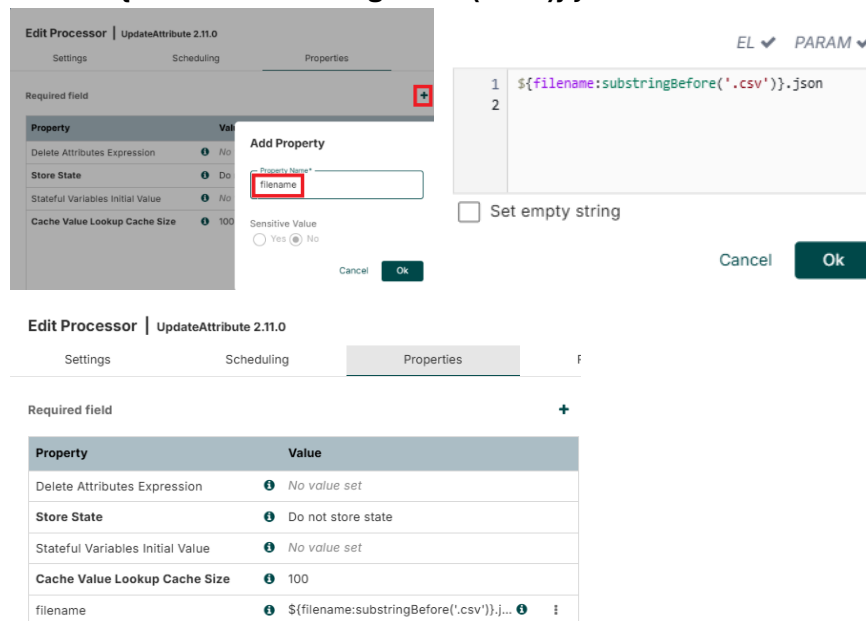


c. Klik bikoitza egin **UpdateAttribute** -en gainean, propietateak aldatzeko.



d. **Properties** atalean:

- Propietate berria gehitu **+** ikurra sakatuz: **filename**
- Balioa: **`${filename:substringBefore('.csv')}.json`**



5. **Output Port** irteera-portua gehitu:



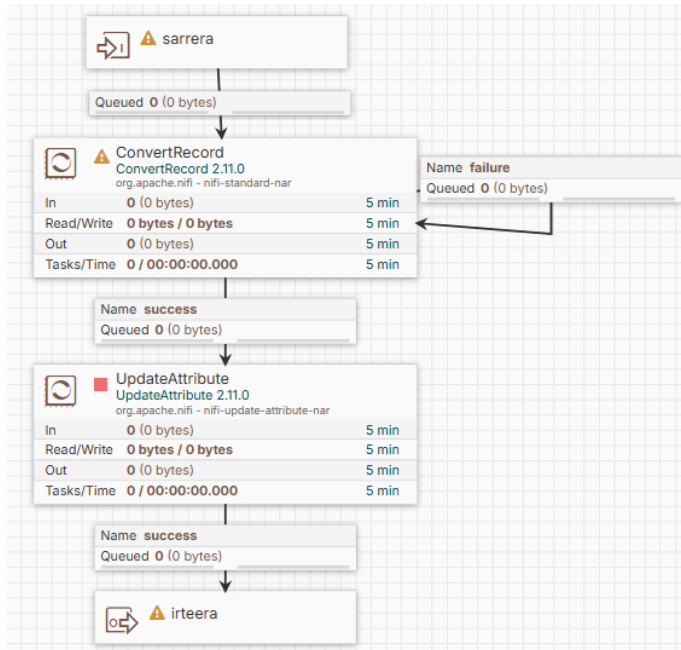
Create New Output Port

Output Port Name*

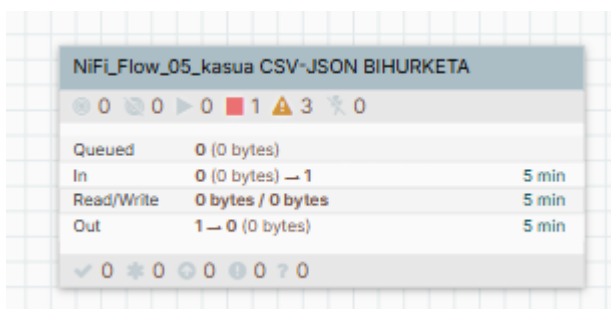
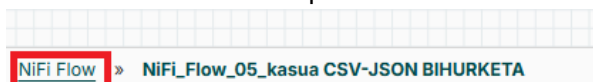
Cancel Add

6. Konexioak ezarri:

- Input Port (sarrera) -> ConvertRecord:** *success* konexioa
- ConvertRecord -> UpdateAttribute:** *success* konexioa
- ConvertRecord -> ConvertRecord:** *failure* konexioa
- UpdateAttribute -> Output Port (irteera):** *success* konexioa



7. Sortutako Process Group-etik irten.



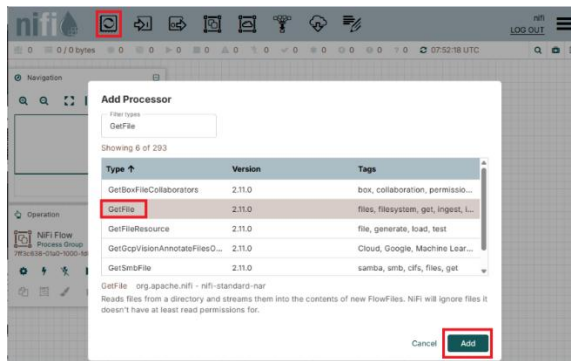
8. **GetFile** prozesadorea:

Gehitu **GetFile** prozesadorea, sarrera-karpetatik fitxategiak eskuratzeko (irakurri eta kendu):

- Processor** arrastatu.



- GetFile** aukeratu eta **Add** sakatu.



c. Klik bikoitza egin **GetFile**-en gainean, propietateak aldatzeko.

	GetFile GetFile 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

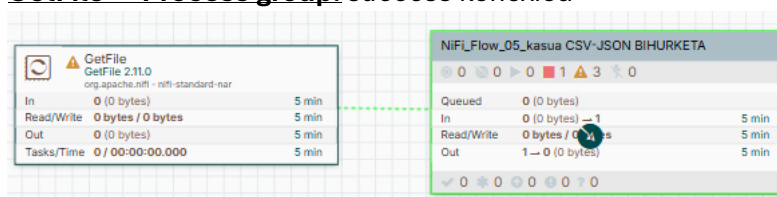
d. **Properties** atalean:

- **Input Directory:** /opt/nifi/ariketak/05-ariketa-csv-json-bihurtu/sarrera
- **Keep Source File:** false
- **File Filter:** [^\.]*
- **Minimum File Age:** 2 sec

Edit Processor GetFile 2.11.0	
Settings	Scheduling
Properties	
Required field +	
Property	Value
Input Directory	/opt/nifi/ariketak/05-ariketa-csv-json...
File Filter	[^\.]*
Path Filter	No value set
Batch Size	10
Keep Source File	false
Recurse Subdirectories	true
Polling Interval	30 sec
Ignore Hidden Files	true
Minimum File Age	2 sec
Maximum File Age	No value set
Minimum File Size	0 B
Maximum File Size	No value set

9. **GetFile** prozesadorea **Process group**-ari konektatu:

a. **GetFile -> Process group:** success konexioa



Create Connection

Details

From Processor
GetFile

Within Group
NiFi Flow

Relationships
☒ success

Settings

To Input
sarrera

Within Group
NiFi_Flow_05_kasua CSV-JSON BIHURKETA

10. **PutFile** prozesadorea:

Gehitu **PutFile** prozesadorea, irteera-karpetan idazteko.

a. **Processor** arrastatu.



b. **PutFile** aukeratu eta **Add** sakatu.

Add Processor

Filter types
putfi

Showing 1 of 293

Type ↑	Version	Tags
PutFile	2.11.0	archive, copy, files, filesystem...

c. Klik bikoitza egin **PutFile**-en gainean, propietateak aldatzeko.

PutFile 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- **Directory:** /opt/nifi/ariketak/05-ariketa-csv-json-bihurtu/irteera
- **Conflict Resolution Strategy:** fail

Edit Processor | PutFile 2.11.0

Settings | Scheduling | **Properties** | R

Required field +

Property	Value
Directory	/opt/nifi/ariketak/05-ariketa-csv-json-...
Conflict Resolution Strategy	fail
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

e. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**

- failure: **terminate**
- success: **terminate**

Edit Processor | PutFile 2.11.0

Settings Scheduling Properties Relationships

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

Files that could not be written to the output directory for some reason are transferred to this relationship

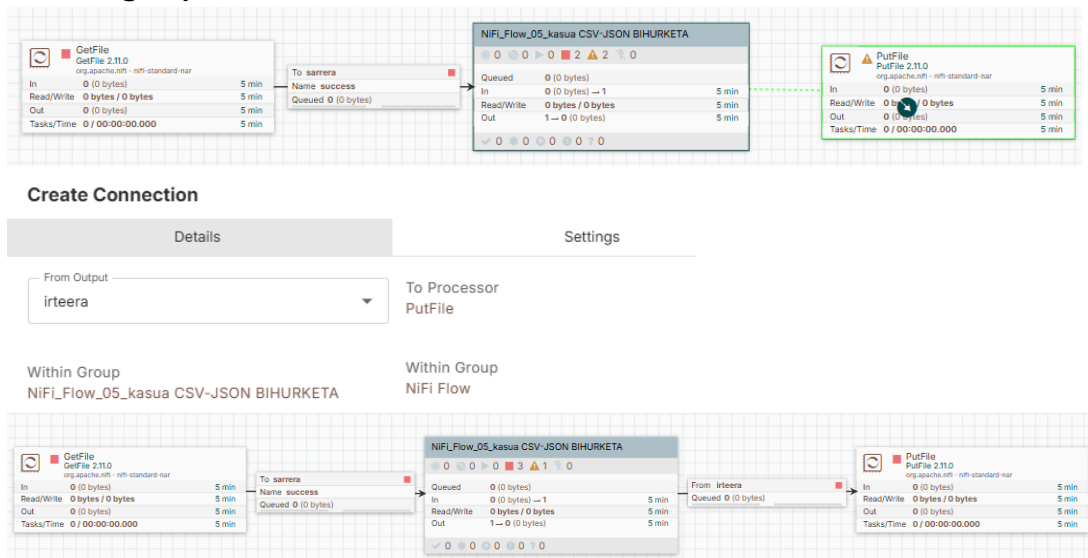
success

☒ terminate ☐ retry

Files that have been successfully written to the output directory are transferred to this relationship

11. Process Group PutFile prozesadoreari konektatu:

a. Process group -> PutFile: success konexioa



12. Proba egiteko:

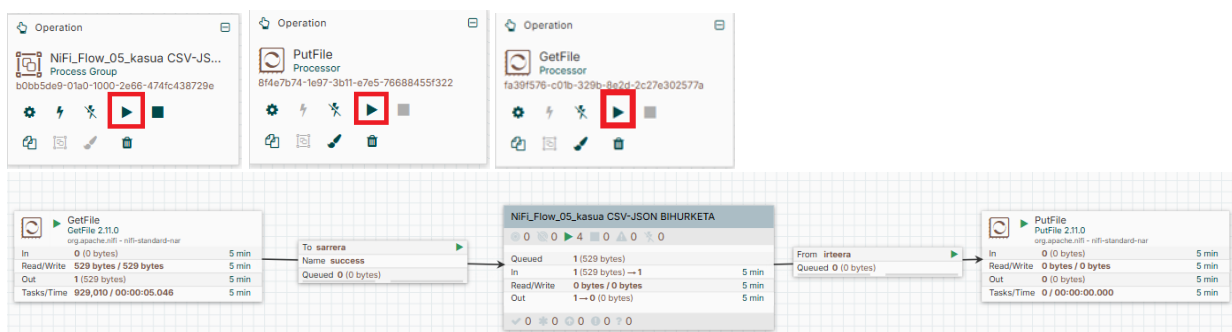
a. **Sarrera**-karpetan .csv fitxategi bat sortu (**salmentak.csv**):

arriketak > 05-arriketa-csv-json-bihurtu > sarrera Buscar en sarrera

Nombre	Fecha de modificación	Tipo	Tamaño
salmentak.csv	13/09/2026 10:53	Archivo CSV	1 KB

Ordenar Ver Detalles

b. Sortutako fluxua exekutatu:



c. Egiaztatu **sarrera**-karpetatik CSV kendu eta **irteera**-karpetara pasatu dela, JSON bihurtuz.

... ariketak > 05-ariketa-csv-json-bihurtu > sarrera

Buscar en sarrera

Ordenar Ver ... Detalles

Nombre	Fecha de modificación	Tipo	Tamaño
Esta carpeta está vacía.			

... ariketak > 05-ariketa-csv-json-bihurtu > irteera

Buscar en irteera

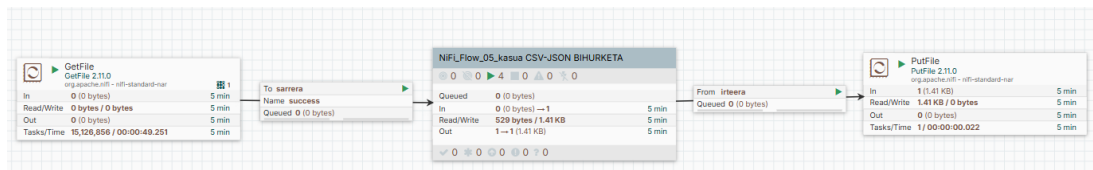
Ordenar Ver ... Detalles

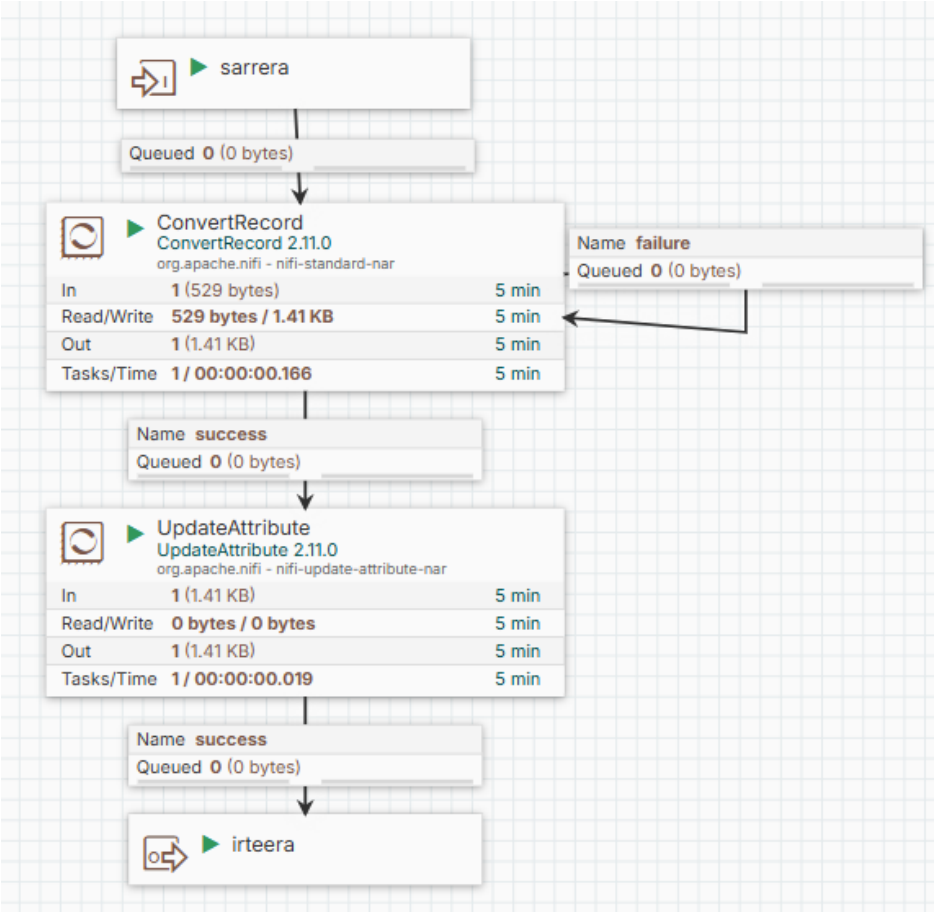
Nombre	Fecha de modificación	Tipo	Tamaño
salmentak.json	17/09/2026 21:40	Archivo JSON	2 KB

salmentak.json

Archivo Editar Ver

```
[{"ProductID": "725", "Date": "1/15/1999", "Zip": "41540", "Units": "1", "Revenue": "115.5", "Country": "Germany"}, {"ProductID": "850", "Date": "2/03/1999", "Zip": "75000", "Units": "3", "Revenue": "245.0", "Country": "France"}, {"ProductID": "425", "Date": "3/21/1999", "Zip": "28013", "Units": "1", "Revenue": "87.25", "Country": "Spain"}, {"ProductID": "725", "Date": "4/12/1999", "Zip": "75008", "Units": "5", "Revenue": "577.5", "Country": "France"}, {"ProductID": "910", "Date": "5/05/1999", "Zip": "10115", "Units": "2", "Revenue": "230.0", "Country": "Germany"}, {"ProductID": "850", "Date": "6/30/1999", "Zip": "69001", "Units": "4", "Revenue": "392.0", "Country": "France"}, {"ProductID": "# Iragazketaren xedea:", "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": "# Country = 'France' ETA Units > 1", "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": "# Helburua: Frantziako salmentak unitate bat baino gehiago", "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": "# duten erregistroak SOILIK pasatzea.", "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": null, "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": "# Bereizlea:", "Date": "(puntu eta koma) - ohikoa Europar formatuan", "Zip": null, "Units": null, "Revenue": null, "Country": null}, {"ProductID": "# Lehen lerroa: GOIBURUA (zutabe-izenak)", "Date": null, "Zip": null, "Units": null, "Revenue": null, "Country": null}]
```





Apache NiFi 6. kasua: MariaDB → MongoDB

Bi datu baseen arteko integrazioa.

MariaDB (datu-base erlazionala) → NiFi → MongoDB (datu-base NoSQL)

Prozesadore berriak:

- **ExecuteSQLRecord** [*DBCPConnectionPool* controller service-a]
- **SplitText** + **PutMongo** [*MongoDBControllerService* controller service-a] (1. aldaera)
- **PutMongoRecord** (2. aldaera)

Bi aldaera:

1. SplitText + PutMongo
2. PutMongoRecord

Datu-iturria: *retail_db* datu-base erlazionala

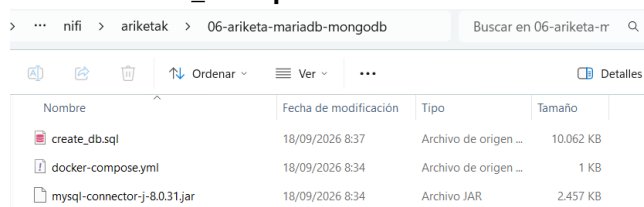
Ingurunea: **Docker compose** erabiliko dugu ingurune osoa komando berean abiarazteko

- **iabd-mysql-nifi**
- **iabd-mongodb-nifi**
- **iabd-nifi**

1. **DOCKER COMPOSE:**

a. Karpeta prestatu:

- **docker-compose.yaml**
- **mysql-connector-j-8.0.31.jar**
- **create_db.sql**



Nombre	Fecha de modificación	Tipo	Tamaño
create_db.sql	18/09/2026 8:37	Archivo de origen ...	10.062 KB
docker-compose.yml	18/09/2026 8:34	Archivo de origen ...	1 KB
mysql-connector-j-8.0.31.jar	18/09/2026 8:34	Archivo JAR	2.457 KB

b. Aurreko nifi container-ak geratu:

docker stop nifi
docker stop mongodb

c. Ingurune berria abiarazi:

- Sartu karpetan, terminala ireki eta exekutatu:
docker compose -p iabd-nifi-mysql-mongodb up -d

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\nifi\ariketak\06-ariketa-mariadb-mongodb> docker compose -p iabd-nifi-mysql-mongodb up -d
✓Image mysql:latest Pulled
✓Image mongo:latest Pulled
✓Network iabd-nifi-mysql-mongodb_default Created
✓Container iabd-mysql-nifi Started
✓Container iabd-mongodb-nifi Started
✓Container iabd-nifi Started
PS C:\nifi\ariketak\06-ariketa-mariadb-mongodb>

```

Docker Desktop-en hiru container hauek agertuko dira:

- iabd-nifi
- iabd-mysql-nifi
- iabd-mongodb-nifi

✓	iabd-nifi-mysql-mong	-	-	-	1.82%	1 minute ago				
●	iabd-mysql-nifi	d83f57d103f2	mysql:latest	3306:3306	0.42%	1 minute ago				
●	iabd-mongodb-nifi	e962875e5877	mongo:latest	27017:27017	0.58%	1 minute ago				
●	iabd-nifi	e67d9a967e39	apache/nifi:latest	8443:8443	0.82%	1 minute ago				

d. Inportatu retail_db datu-basea terminaletik:

cmd

docker exec -i iabd-mysql-nifi mysql -uiabd -piabd retail_db < create_db.sql

```

PS C:\nifi\ariketak\06-ariketa-mariadb-mongodb> cmd
Microsoft Windows [Version 10.0.26280.9457]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\nifi\ariketak\06-ariketa-mariadb-mongodb>docker exec -i iabd-mysql-nifi mysql -h 0.0.0.0 -P 3306 -uiabd -piabd retail_db < create_db.sql
mysql: [Warning] Using a password on the command line interface can be insecure.

C:\nifi\ariketak\06-ariketa-mariadb-mongodb>

```

e. Ondo kargatu dela egiaztatzeko:

docker exec -it iabd-mysql-nifi mysql -uiabd -piabd retail_db

```

C:\nifi\ariketak\06-ariketa-mariadb-mongodb>docker exec -it iabd-mysql-nifi mysql -uiabd -piabd retail_db
mysql: [Warning] Using a password on the command line interface can be insecure.
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 26.7.0 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>

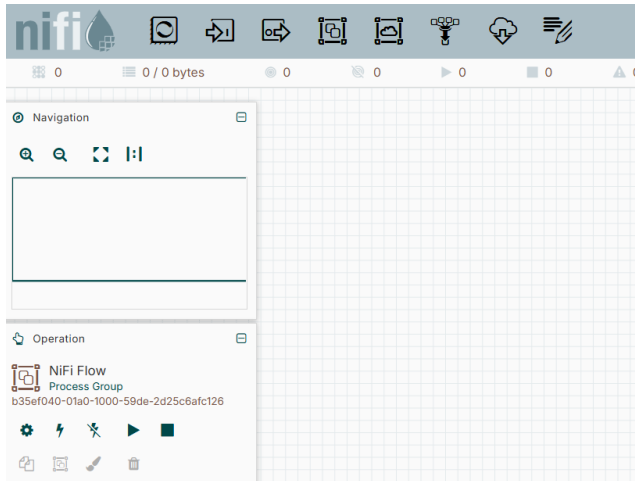
```

Barruan gaudela:

show tables;

```
mysql> show tables;
+-----+
| Tables_in_retail_db |
+-----+
| categories           |
| customers            |
| departments          |
| order_items          |
| orders              |
| products             |
+-----+
6 rows in set (0.001 sec)
```

2. **NiFi ireki nabigatzailean (<https://localhost:8443/nifi>):**



3. **ExecuteSQLRecord prozesadorea**

	ExecuteSQLRecord ExecuteSQLRecord 2.11.0 org.apache.nifi - nifi-standard-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties atalean:**

- **Database Connection Pooling Service:** DBCPConnectionPool
DBCConnectionPool konfiguratu:
 - **Database Connection URL:** jdbc:mysql://iabd-mysql-nifi/retail_db
 - **Database Driver Class Name:** com.mysql.cj.jdbc.Driver
 - **Database Driver Location:** /opt/mysql-connector-j-8.0.31.jar
 - **Database User:** iabd
 - **Password:** iabd

Edit Controller Service

Property	Value
Database Connection URL	jdbc:mysql://labd-mysql-nifi/retail_db
Database Driver Class Name	com.mysql.cj.jdbc.Driver
Database Driver Locations	/opt/nifi/ariketak/06-ariкета-mariadb...
Kerberos User Service	No value set
Database User	labd
Password Source	Password
Password	Sensitive value set
Max Wait Time	500 millis
Max Total Connections	8
Validation Query	No value set

- **SQL Query:** `SELECT * FROM customers`
 - **Record Writer:** `JsonRecordSetWriter` (Output Grouping: 'One Line per Object')
- JsonRecordSetWriter konfiguratu:
- **Output Grouping: One Line per Object**

Edit Controller Service

Property	Value
Schema Write Strategy	Do Not Write Schema
Schema Cache	No value set
Schema Access Strategy	Inherit Record Schema
Date Format	No value set
Time Format	No value set
Timestamp Format	No value set
Pretty Print JSON	false
Suppress Null Values	Never Suppress
Allow Scientific Notation	false
Output Grouping	One Line Per Object

Edit Processor | ExecuteSQLRecord 2.11.0

Property	Value
Database Connection Pooling Ser...	DBCPConnectionPool
SQL Pre-Query	No value set
SQL Query	SELECT * FROM customers
SQL Post-Query	No value set
Max Wait Time	0 seconds
Record Writer	JsonRecordSetWriter
Normalize Table/Column Names	false
Use Avro Logical Types	false
Default Decimal Precision	10
Default Decimal Scale	0

b. **Scheduling** atalean:

Run schedule: 0 sec (eskuz exekutatzeko)

Edit Processor | ExecuteSQLRecord 2.11.0

Settings	Scheduling	Properties
Scheduling Strategy ⓘ*		
Timer driven		
Concurrent Tasks ⓘ*	Run Schedule ⓘ*	
1	0 sec	
Execution ⓘ*		
All nodes		

4. **SplitText** prozesadorea:

FF-aren edukia JSON dokumentu osoa denez, beharrezkoa da SplitText prozesadorea (lerro bakoitzeko FF bat).

	 SplitText SplitText 2.11.0 org.apache.nifi - nifi-standard-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties** atalean:

- **Line Split Count: 1** (lerro bakoitza FF independente)
- **Header Line Count: 0** (JSON-en ez dago goibururik)

Edit Processor | SplitText 2.11.0

Settings	Scheduling	Properties	Ⓜ												
Required field +															
<table><thead><tr><th>Property</th><th>Value</th></tr></thead><tbody><tr><td>Line Split Count</td><td>1</td></tr><tr><td>Maximum Fragment Size</td><td>No value set</td></tr><tr><td>Header Line Count</td><td>0</td></tr><tr><td>Header Line Marker Characters</td><td>No value set</td></tr><tr><td>Remove Trailing Newlines</td><td>true</td></tr></tbody></table>				Property	Value	Line Split Count	1	Maximum Fragment Size	No value set	Header Line Count	0	Header Line Marker Characters	No value set	Remove Trailing Newlines	true
Property	Value														
Line Split Count	1														
Maximum Fragment Size	No value set														
Header Line Count	0														
Header Line Marker Characters	No value set														
Remove Trailing Newlines	true														

5. **PutMongo** prozesadorea:

	PutMongo PutMongo 2.11.0 org.apache.nifi - nifi-mongodb-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties** atalean:

- **Client Service: MongoDBControllerService**
MongoDBControllerService konfiguratu:
 - **MongoDB URI: mongodb://iabd-mongodb-nifi:27017**

Edit Controller Service

Settings	Properties												
Required field +													
<table><thead><tr><th>Property</th><th>Value</th></tr></thead><tbody><tr><td>Mongo URI</td><td>mongodb://iabd-mongodb-nifi</td></tr><tr><td>Database User</td><td>No value set</td></tr><tr><td>Password</td><td>No value set</td></tr><tr><td>SSL Context Service</td><td>No value set ⓘ</td></tr><tr><td>Write Concern</td><td>ACKNOWLEDGED</td></tr></tbody></table>		Property	Value	Mongo URI	mongodb://iabd-mongodb-nifi	Database User	No value set	Password	No value set	SSL Context Service	No value set ⓘ	Write Concern	ACKNOWLEDGED
Property	Value												
Mongo URI	mongodb://iabd-mongodb-nifi												
Database User	No value set												
Password	No value set												
SSL Context Service	No value set ⓘ												
Write Concern	ACKNOWLEDGED												

- **Mongo Database Name: iabd**
- **Mongo Collection Name: 6kasua**
- **Mode: insert**

Edit Processor | PutMongo 2.11.0

Settings Scheduling Properties

Required field +

Property	Value
Client Service	MongoDBControllerService
Mongo Database Name	iabd
Mongo Collection Name	6kasua
Mode	insert
Character Set	UTF-8

6. Konexioak ezarri:

a. Prozesadoreak elkartu:

- ExecuteSQLRecord → SplitText: success
- SplitText → PutMongo: splits

b. Gainerako harremanak AUTO-TERMINATE:

- ExecuteSQLRecord:
 - Automatically Terminate/Retry:
failure: terminate

Edit Processor | ExecuteSQLRecord 2.11.0

Settings Scheduling Properties Relationships

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

SQL query execution failed. Incoming FlowFile will be penalized and routed to this relationship

success

☐ terminate ☐ retry

Successfully created FlowFile from SQL query result set.

- SplitText:
 - Automatically Terminate/Retry:
failure: terminate
original: terminate

Edit Processor | SplitText 2.11.0

Settings Scheduling Properties Relationships

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

If a file cannot be split for some reason, the original file will be routed to this destination and nothing will be routed elsewhere

original

☒ terminate ☐ retry

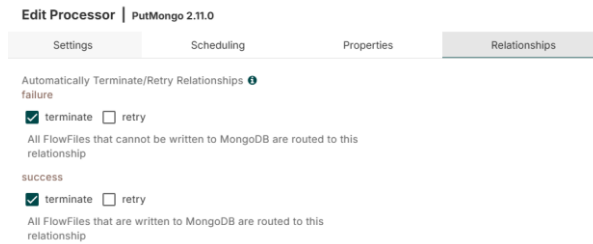
The original input file will be routed to this destination when it has been successfully split into 1 or more files

splits

☐ terminate ☐ retry

The split files will be routed to this destination when an input file is successfully split into 1 or more split files

- PutMongo:
 - Automatically Terminate/Retry:
failure: terminate
success: terminate



7. Egiaztatu funtzionatzen duela:

- a. MongoDBko collection-a hutsik dagoela ziurtatu:

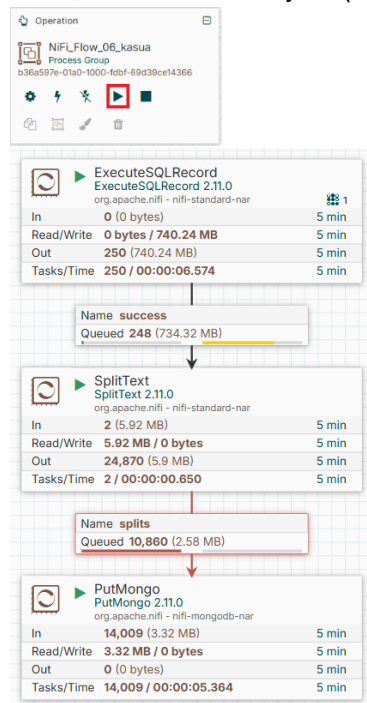
docker exec -it iabd-mongodb-nifi mongosh

Eta gero:

use iabd

show collections

- b. NiFi-n fluxua martxan jarri (**Start**).



- c. Itxaron pixka bat eta berriro geratu (**Stop**).

Run Schedule = 0 sec denez, ExecuteSQLRecord martxan uzten badugu, SQL kontsulta behin eta berriro exekutatu da.

- d. MongoDB-n egiaztatzeko, terminalean exekutatu:

docker exec -it iabd-mongodb-nifi mongosh

```
C:\nifi\ariketak\06-ariketa-mariadb-mongodb>docker exec -it iabd-mongodb-nifi mongosh
Current Mongosh Log ID: 6aacf850e9fc5e79b0766e14
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeo
utMS=2000&appName=mongosh+2.11.1
Using MongoDB:      8.3.11
Using Mongosh:      2.11.1

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-09-18T07:15:25.714+00:00: Using the XFS filesystem is strongly recommended with the W
iredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
  2026-09-18T07:15:25.812+00:00: Access control is not enabled for the database. Read and wr
ite access to data and configuration is unrestricted
  2026-09-18T07:15:25.812+00:00: For customers running the current memory allocator, we sugg
est changing the contents of the following sysfsFile
  2026-09-18T07:15:25.812+00:00: For customers running the current memory allocator, we sugg
est changing the contents of the following sysfsFile
  2026-09-18T07:15:25.812+00:00: We suggest setting the contents of sysfsFile to 0.
  2026-09-18T07:15:25.812+00:00: vm.max_map_count is too low
  2026-09-18T07:15:25.812+00:00: We suggest setting swappiness to 0 or 1, as swapping can ca
use performance problems.
-----

test>
```

Eta gero:

use iabd

```
test> use iabd
switched to db iabd
```

show collections

```
iabd> show collections
6kasua
```

db.getCollection("6kasua").find().limit(5)

```
iabd> db.getCollection("6kasua").find().limit(5)
[
  {
    _id: ObjectId('6aacf7f2de7ce7be6fbc408'),
    customer_id: 1,
    customer_fname: 'Richard',
    customer_lname: 'Hernandez',
    customer_email: 'XXXXXXXXXX',
    customer_password: 'XXXXXXXXXX',
    customer_street: '6303 Heather Plaza',
    customer_city: 'Brownsville',
    customer_state: 'TX',
    customer_zipcode: '78521'
  },
  {
    _id: ObjectId('6aacf7f2de7ce7be6fbc409'),
    customer_id: 2,
    customer_fname: 'Mary',
    customer_lname: 'Barrett',
    customer_email: 'XXXXXXXXXX',
    customer_password: 'XXXXXXXXXX',
    customer_street: '9526 Noble Embers Ridge',
    customer_city: 'Littleton',
    customer_state: 'CO',
    customer_zipcode: '80126'
  },
  {
    _id: ObjectId('6aacf7f2de7ce7be6fbc40a'),
    customer_id: 3,
    customer_fname: 'John',
    customer_lname: 'Doe',
    customer_email: 'XXXXXXXXXX',
    customer_password: 'XXXXXXXXXX',
    customer_street: '1234 Main St',
    customer_city: 'New York',
    customer_state: 'NY',
    customer_zipcode: '10001'
  },
  {
    _id: ObjectId('6aacf7f2de7ce7be6fbc40b'),
    customer_id: 4,
    customer_fname: 'Jane',
    customer_lname: 'Smith',
    customer_email: 'XXXXXXXXXX',
    customer_password: 'XXXXXXXXXX',
    customer_street: '5678 Elm St',
    customer_city: 'Los Angeles',
    customer_state: 'CA',
    customer_zipcode: '90001'
  },
  {
    _id: ObjectId('6aacf7f2de7ce7be6fbc40c'),
    customer_id: 5,
    customer_fname: 'Mike',
    customer_lname: 'Johnson',
    customer_email: 'XXXXXXXXXX',
    customer_password: 'XXXXXXXXXX',
    customer_street: '9012 Oak St',
    customer_city: 'Chicago',
    customer_state: 'IL',
    customer_zipcode: '60601'
  }
]
```

db.getCollection("6kasua").countDocuments()

```
iabd> db.getCollection("6kasua").countDocuments()
39100
iabd>
```

6. kasuaren 2. aldaera: PutMongoRecord

SplitText eta PutMongo erabili ordeztuz, **PutMongoRecord** prozesadorea erabiliko dugu kasu honetan.

Abantaila: SplitText kenduta, FF batean dokumentu anitz sartzen ditu eta BATCH INSERT bakar batean MongoDB-ra pasatzen dira (askoz azkarragoa). Modu honetan errendimendu hobea lortzen da (batch bidezko txertaketa, memoria txikiagoa, FF gutxiago). Aldaera honekin lortzen dugu produkzio-praktika gomendagarria datu-base integrazioan.

ExecuteSqlRecord prozesadoreak NDJSON (*Newline Delimited JSON*, hau da, lerro bakoitzean JSON objektu independente bat) sortzen du eta **PutMongoRecord** prozesadoreak NDJSON irakurtzen du **JsonTreeReader** kontrol-zerbitzuaren bidez.

1. **PutMongoRecord** prozesadorea:

	PutMongoRecord PutMongoRecord 2.11.0 org.apache.nifi - nifi-mongodb-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties** atalean:

- **Client Service:** MongoDBControllerService
- **Mongo Database Name:** iabd
- **Mongo Collection Name:** 6kasua-record
- **Record Reader:** JsonTreeReader

Edit Processor PutMongoRecord 2.11.0		
Settings	Scheduling	Properties
Required field +		
Property	Value	
Client Service	MongoDBControllerService	
Mongo Database Name	iabd	
Mongo Collection Name	6kasua-record	
Record Reader	JsonTreeReader	
Batch Size	100	
Ordered	False	
Bypass Validation	False	
Update Key Fields	No value set	

2. **SplitText** eta **PutMongo** prozesadoreak kendu.

3. **Konexioak ezarri:**

a. Prozesadoreak elkartu:

- ExecuteSQLRecord → PutMongoRecord: success

b. Gainerako harremanak AUTO-TERMINATE:

- ExecuteSQLRecord:
 - Automatically Terminate/Retry:
failure: terminate

Edit Processor | ExecuteSQLRecord 2.11.0

Settings	Scheduling	Properties	Relationships
Automatically Terminate/Retry Relationships ⓘ failure ☒ terminate ☐ retry SQL query execution failed. Incoming FlowFile will be penalized and routed to this relationship success ☐ terminate ☐ retry Successfully created FlowFile from SQL query result set.			

- PutMongoRecord:
 - Automatically Terminate/Retry:
 - failure: terminate**
 - success: terminate**

Edit Processor | PutMongoRecord 2.11.0

Settings	Scheduling	Properties	Relationships
Automatically Terminate/Retry Relationships ⓘ failure ☒ terminate ☐ retry All FlowFiles that cannot be written to MongoDB are routed to this relationship success ☒ terminate ☐ retry All FlowFiles that are written to MongoDB are routed to this relationship			

4. Egiaztatu funtzionatzen duela:

- MongoDBko collection-a hutsik dagoela ziurtatu:

docker exec -it iabd-mongodb-nifi mongosh

```

mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.11.1
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\System32> docker exec -it iabd-mongodb-nifi mongosh
Current Mongosh Log ID: 6aa66a11bdf5e161701a
Connecting to:   mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.11.1
Using MongoDB:  8.3.11
Using Mongosh:  2.11.1

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-19T10:17:03.811+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-19T10:17:04.092+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-19T10:17:04.092+00:00: vm.max_map_count is too low
2026-09-19T10:17:04.092+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
test> |

```

Eta gero egiaztatu 6kasua-record oraindik ez daukagula sortuta:

use iabd

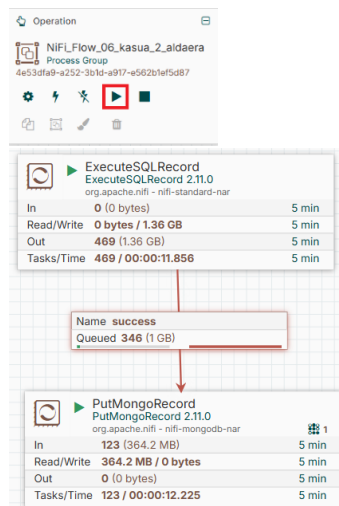
show collections

```

test> use iabd
switched to db iabd
iabd> show collections
6kasua

```

- NiFi-n fluxua martxan jarri (**Start**).



- c. Itxaron pixka bat eta berriro geratu (**Stop**).
Run Schedule = 0 sec denez, ExecuteSQLRecord martxan uzten badugu, SQL kontsulta behin eta berriro exekutatu da.
- d. MongoDB-n egiaztatzeko, terminalean exekutatu:

docker exec -it iabd-mongodb-nifi mongosh

```
PS C:\WINDOWS\System32> docker exec -it iabd-mongodb-nifi mongosh
Current Mongosh Log ID: 6aa6eb3939426d6f1f6258dd
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.11.1
Using MongoDB: 8.3.11
Using Mongosh: 2.11.1

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-19T10:17:03.811+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-19T10:17:04.092+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-19T10:17:04.092+00:00: vm.max_map_count is too low
2026-09-19T10:17:04.092+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
```

Eta gero:

use iabd

```
test> use iabd
switched to db iabd
```

show collections

```
iabd> show collections
6kasua
6kasua-record
```

db.getCollection("6kasua-record").find().limit(5)

```

mongosh mongodb://127.0.0.1
iabd> db.getCollection("6kasua-record").find().limit(5)
[
  {
    _id: ObjectId('6aae6ae8e0e480ff35750525'),
    customer_id: 1,
    customer_fname: 'Richard',
    customer_lname: 'Hernandez',
    customer_email: 'XXXXXXXXX',
    customer_password: 'XXXXXXXXX',
    customer_street: '6303 Heather Plaza',
    customer_city: 'Brownsville',
    customer_state: 'TX',
    customer_zipcode: '78521'
  },
  {
    _id: ObjectId('6aae6ae8e0e480ff35750526'),
    customer_id: 2,
    customer_fname: 'Mary',
    customer_lname: 'Barrett',
    customer_email: 'XXXXXXXXX',
    customer_password: 'XXXXXXXXX',
    customer_street: '9526 Noble Embers Ridge',
    customer_city: 'Littleton',
    customer_state: 'CO',
    customer_zipcode: '80126'
  },
  {
    _id: ObjectId('6aae6ae8e0e480ff35750527'),
    customer_id: 3,
    customer_fname: 'Ann',

```

db.getCollection("6kasua-record").countDocuments()

```

iabd> db.getCollection("6kasua-record").countDocuments()
2524305
iabd>

```

Apache NiFi 7. kasua: AEMET Datu-lakua

Helburua: Produkzio-ingurune batean aurki dezakegun datu-arkitektura oso bat ikustea. AEMET-eko datuekin datu-laku bat eraikiko dugu, Medallion Architecture eredua jarraituz: Bronze, Silver, Gold. Gainera, datuak bi helbururekin gordeko ditugu:

- Fitxategi sisteman: historikoa eta analitika egiteko
- MongoDB-n: aplikazioek datuak azkar kontsultatu ahal izateko.

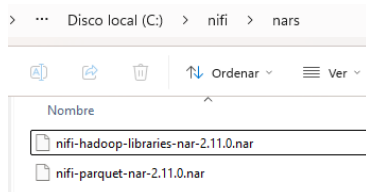
Fluxua:

- **BRONZE: InvokeHTTP → UpdateAttribute → PutFile**
JSON gordina fitxategi-sisteman: */opt/nifi/ariketak/07-ariketa.../bronze*
- **SILVER: EvaluateJSONPath → AttributesToJSON**
→ Silver JSONa → *(fitxategi-sistema)* **PutFile** + *(MongoDB)* **PutMongo** [paraleloan]
- **GOLD: MergeContent → UpdateAttribute → QueryRecord**
→ Gold Parquet agregatua → *(fitxategi-sistema)* **PutFile** + *(MongoDB)* **PutMongoRecord**

Praktikarako beharrezko konfigurazio gehigarriak:

Praktika honekin hasi aurretik NiFi Docker instalazioari bi NAT gehitu behar zaizkio: **nifi-parquet-nar** eta **nifi-hadoop-libraries-nar**.

- Geratu nifi kontenedorea.
- Deskargatu 2.11 bertsio zehatzak:
[nifi-parquet-nar-2.11.0.nar](#)
[nifi-hadoop-libraries-nar-2.11.0.nar](#)
- Sortu karpeta bat Windows-en:
C:\nifi\nars
Eta bertan kopiatu deskargatutako bertsioak.



- Kopiatu NAR-ak NiFi kontenedorean:
docker cp "C:\nifi\nars\nifi-hadoop-libraries-nar-2.11.0.nar" iabd-nifi:/opt/nifi/nifi-current/lib/

```
PS C:\WINDOWS\System32> docker cp "C:\nifi\nars\nifi-hadoop-libraries-nar-2.11.0.nar" iabd-nifi:/opt/nifi/nifi-current/lib/
Successfully copied 59.5MB (transferred 59.5MB) to iabd-nifi:/opt/nifi/nifi-current/lib/
```

docker cp "C:\nifi\nars\nifi-parquet-nar-2.11.0.nar" iabd-nifi:/opt/nifi/nifi-current/lib/

```
PS C:\WINDOWS\System32> docker cp "C:\nifi\nars\nifi-parquet-nar-2.11.0.nar" iabd-nifi:/opt/nifi/nifi-current/lib/
Successfully copied 15.1MB (transferred 15.1MB) to iabd-nifi:/opt/nifi/nifi-current/lib/
```

- Abiarazi berriro NiFi:
docker start iabd-nifi

- Egiaztatu barruan daudela:
docker exec iabd-nifi sh -c "ls -lh /opt/nifi/nifi-current/lib/*parquet* /opt/nifi/nifi-current/lib/*hadoop-libraries*"

```
PS C:\WINDOWS\System32> docker exec iabd-nifi sh -c "ls -lh /opt/nifi/nifi-current/lib/*parquet* /opt/nifi/nifi-current/lib/*hadoop-libraries*"
-rwxr-xr-x 1 root root 57M Sep 20 15:38 /opt/nifi/nifi-current/lib/nifi-hadoop-libraries-nar-2.11.0.nar
-rw-r--r-- 1 nifi nifi 24M Jul 30 21:11 /opt/nifi/nifi-current/lib/nifi-iceberg-parquet-writer-nar-2.11.0.nar
-rwxr-xr-x 1 root root 15M Sep 20 15:39 /opt/nifi/nifi-current/lib/nifi-parquet-nar-2.11.0.nar
```

Orain gutxienez hauek ikusi beharko dira:

nifi-hadoop-libraries-nar-2.11.0.nar

nifi-parquet-nar-2.11.0.nar

1. InvokeHTTP prozesadorea:

	InvokeHTTP REST AEMET InvokeHTTP 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

a. Propiedades atalean:

- **HTTP Method:** GET
- **HTTP URL:** https://api.el-tiempo.net/json/v3/provincias/03/municipios/03065?utm_source=chatgpt.com

Edit Processor | InvokeHTTP 2.11.0

SettingsSchedulingProperties

Required field

Property	Value
HTTP Method	GET
HTTP URL	https://www.el-tiempo.net/api/json/v...
HTTP/2 Disabled	False
SSL Context Service	No value set
Connection Timeout	5 secs
Socket Read Timeout	15 secs
Socket Write Timeout	15 secs
Socket Idle Timeout	5 mins
Socket Idle Connections	5
Proxy Configuration Service	No value set
Request OAuth2 Access Token Pro...	No value set

Edit Processor | InvokeHTTP 2.11.0

SettingsSchedulingProperties

Name*

InvokeHTTP REST AEMET

Id

bd7430a3-01a0-1000-7564-dd7e0d010662

Type

InvokeHTTP 2.11.0

Bundle

org.apache.nifi - nifi-standard-nar

Penalty Duration *

30 sec

Yield Duration *

1 sec

Bulletin Level *

WARN

Edit Processor | InvokeHTTP 2.11.0

SettingsSchedulingProperties

Scheduling Strategy *

Timer driven

Concurrent Tasks *

1

Run Schedule *

60 sec

Execution *

All nodes

2. UpdateAttribute prozesadorea:

	UpdateAttribute fecha / filename UpdateAttribute 2.11.0 org.apache.nifi - nifi-update-attribute-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

a. **Propiedades** atalean:

- Propietate berria: **fecha**: `${now():format('yyyy-MM-dd_HH-mm-ss')}`
- Propietate berria: **filename**: `${now():format('yyyy-MM-dd_HH-mm-ss')}.json`

Edit Processor | UpdateAttribute 2.11.0

Settings | Scheduling | **Properties** | R

Required field +

Property	Value
Delete Attributes Expression	No value set
Store State	Do not store state
Stateful Variables Initial Value	No value set
Cache Value Lookup Cache Size	100
fecha	<code>\${now():format('yyyy-MM-dd_HH:mm:ss')}</code>
filename	<code>\${now():format('yyyy-MM-dd_HH-mm-ss')}</code>

Name* UpdateAttribute fecha / filename

Id bd7e5dfe-01a0-1000-4283-42ffa157fd23

Type UpdateAttribute 2.11.0

Bundle org.apache.nifi - nifi-update-attribute-nar

Penalty Duration 30 sec Yield Duration 1 sec

Bulletin Level WARN

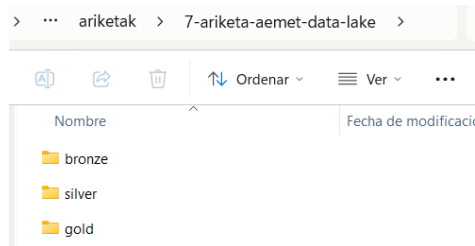
3. **PutFile** prozesadorea:

PutFile 2.11.0

org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. Windows-en karpetak prestatu:



b. **Propiedades** atalean:

- **Directory**: `/opt/nifi/ariketak/07-ariketa-aemet-data-lake/bronze`
- **Conflict Resolution Strategy**: `replace`

Edit Processor | PutFile 2.11.0

Settings | Scheduling | **Properties** | R

Required field +

Property	Value
Directory	<code>/opt/nifi/ariketak/7-ariketa-aemet-data-lake/bronze</code>
Conflict Resolution Strategy	<code>replace</code>
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

Name* PutFile Bronze / RAW

Id bds230b5-01a0-1000-0e78-d5da0c3872ef

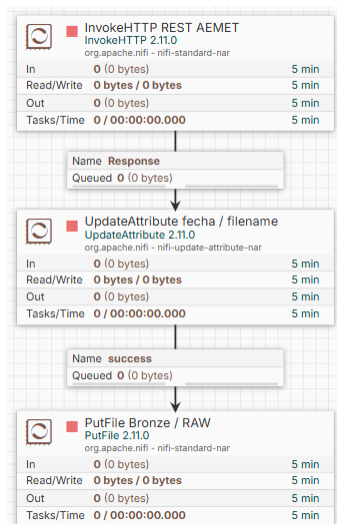
Type PutFile 2.11.0

Bundle org.apache.nifi - nifi-standard-nar

Penalty Duration 30 sec Yield Duration 1 sec

Bulletin Level WARN

4. **Konexioak ezarri:**



5. EvaluateJSONPath prozesadorea:

	EvaluateJSONPath EvaluateJSONPath 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

a. Propiedades atalean:

- **Destination:** flowfile-attribute
- Propietate berria: **hiria:** \$.municipio.NOMBRE
- Propietate berria: **temperatura:** \$.temperatura_actual
- Propietate berria: **hezetasuna:** \$.humedad

Edit Processor EvaluateJSONPath 2.11.0	
Settings	Scheduling
Properties	
Required field +	
Property	Value
Destination	flowfile-attribute
Return Type	auto-detect
Path Not Found Behavior	ignore
Null Value Representation	empty string
Max String Length	20 MB
hezetasuna	\$.humedad
hiria	\$.municipio.NOMBRE
temperatura	\$.temperatura_actual

6. AttributesToJSON prozesadorea:

	AttributesToJSON AttributesToJSON 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

a. Propiedades atalean:

- **Attributes List:** fecha,hiria,temperatura,hezetasuna
- **Destination:** flowfile-content

- **Include Core Attributes: false**

Edit Processor | AttributesToJSON 2.11.0

Settings Scheduling **Properties**

Required field +

Property	Value
Attributes List	fecha,hiria,temperatura,hezetasuna
Attributes Regular Expression	No value set
Destination	flowfile-content
Include Core Attributes	false
Null Value	false
JSON Handling Strategy	Escaped
Pretty Print	false

7. **PutFile prozesadorea:**

 PutFile Silver 1 min
PutFile 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Propiedades** atalean:

- **Directory: /opt/nifi/ariketak/07-ariketa-aemet-data-lake/silver**
- **Conflict Resolution Strategy: replace**

Edit Processor | PutFile 2.11.0

Settings Scheduling **Properties**

Required field +

Property	Value
Directory	/opt/nifi/ariketak/7-ariketa
Conflict Resolution Strategy	replace
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

Edit Processor | PutFile 2.11.0

Settings Scheduling **Property**

Name*
PutFile Silver 1 min

Id
bf389489-01a0-1000-ce99-7722a7a28f87

Type
PutFile 2.11.0

Bundle
org.apache.nifi - nifi-standard-nar

Penalty Duration*
30 sec

Yield Duration*
1 sec

Bulletin Level*
WARN

8. **PutMongo prozesadorea:**

 PutMongo 7-kasua Silver
PutMongo 2.11.0
org.apache.nifi - nifi-mongodb-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties** atalean:

- **Client Service: MongoDBControllerService**
MongoDBControllerService konfiguratu eta abilitatu (Enable):
 - **MongoDB URI: mongodb://iabd-mongodb-nifi:27017**

Edit Controller Service

Settings Properties

Required field +

Property	Value
Mongo URI	mongodb://iabd-mongodb-nifi
Database User	No value set
Password	No value set
SSL Context Service	No value set
Write Concern	ACKNOWLEDGED

- **Mongo Database Name:** iabd
- **Mongo Collection Name:** 7kasua-silver
- **Mode:** insert

Edit Processor | PutMongo 2.11.0

Settings Scheduling Properties

Required field +

Property	Value
Client Service	MongoDBControllerService
Mongo Database Name	iabd
Mongo Collection Name	7kasua-silver
Mode	insert
Character Set	UTF-8

Name* PutMongo 7-kasua Silver

Id bf3db375-01a0-1000-2fca-220f285ee53

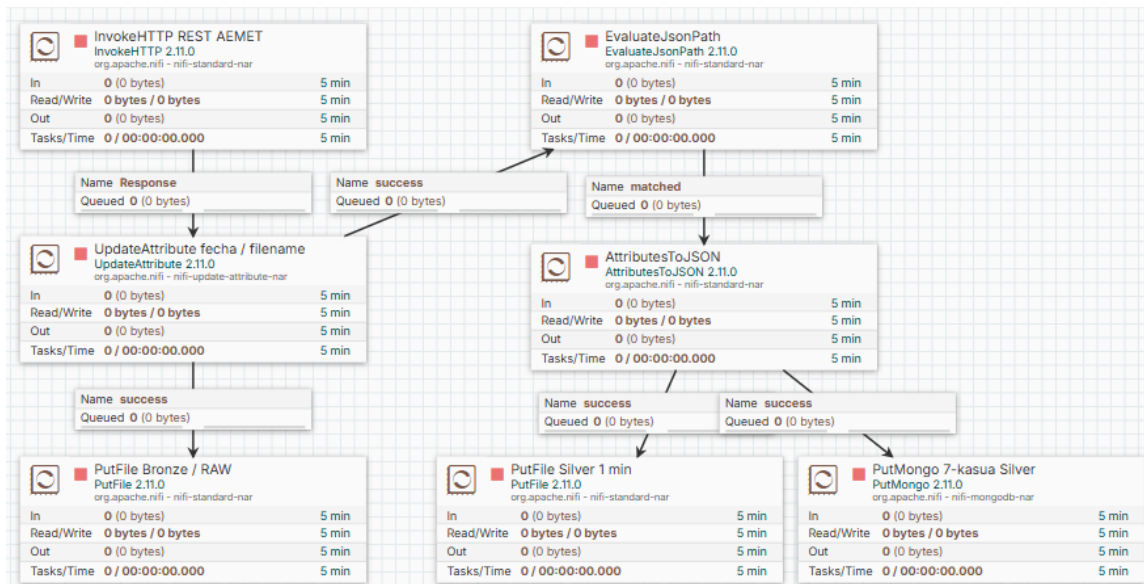
Type PutMongo 2.11.0

Bundle org.apache.nifi - nifi-mongodb-nar

Penalty Duration 30 sec Yield Duration 1 sec

Bulletin Level WARN

9. Konexioak ezarri:



10. MergeContent prozesadorea:

MergeContent 10 FF

MergeContent 2.11.0

org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. Propiedades atalean:

- **Directory:** Bin-Packing Algorithm
- **Merge Format:** Binary Concatenation
- **Minimum Number of Entries:** 10

- **Maximum Number of Entries: 10**
- **Delimiter Strategy: Text**
- **Demarcator: \n**

Edit Processor | MergeContent 2.11.0

Settings
Scheduling
Properties

Required field

Property	Value
Merge Strategy	Bin Packing Algorithm
Merge Format	Binary Concatenation
Attribute Strategy	Keep Only Common Attributes
Correlation Attribute Name	No value set
Minimum Number of Entries	10
Maximum Number of Entries	10
Minimum Group Size	0 B
Maximum Group Size	No value set
Bin Termination Check	No value set
Max Bin Age	No value set
Maximum Number of Bins	5
Delimiter Strategy	Text
Header	No value set
Footer	No value set
Demarcator	\n

Edit Processor | MergeContent 2.11.0

Settings
Scheduling
Properties

Name*

MergeContent 10 FF

Id

bf4b3992-01a0-1000-55a6-744c174e7d6f

Type

MergeContent 2.11.0

Bundle

org.apache.nifi - nifi-standard-nar

Penalty Duration*

30 sec

Yield Duration*

1 sec

Bulletin Level*

WARN

11. UpdateAttribute prozesadorea:

UpdateAttribute
UpdateAttribute 2.11.0
org.apache.nifi - nifi-update-attribute-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. Propiedades atalean:

- Propietate berria: **filename: \${now():format('yyyy-MM-dd_HH-mm-ss')}.parquet**

Edit Processor | UpdateAttribute 2.11.0

Settings
Scheduling
Properties

Required field

Property	Value
Delete Attributes Expression	No value set
Store State	Do not store state
Stateful Variables Initial Value	No value set
Cache Value Lookup Cache Size	100
filename	\${now():format('yyyy-MM-dd_HH-mm-ss')}

12. QueryRecord prozesadorea:

QueryRecord
QueryRecord 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. Properties atalean:

- **Record Reader: JsonTreeReader**
JsonTreeReader konfiguratu eta abilitatu (*Enable*):
- **Timestamp Format: yyyy-MM-dd HH:mm:ss**

Edit Controller Service	
Settings	Properties
Required field +	
Property	Value
Schema Access Strategy	Infer Schema
Schema Inference Cache	No value set
Starting Field Strategy	Root Node
Max String Length	20 MB
Parsing Strategy	Standard
Date Format	No value set
Time Format	No value set
Timestamp Format	yyyy-MM-dd HH:mm:ss

- **Record Writer: ParquetRecordSetWriter** (*enable*)
- **Include Zero Record FlowFiles: false**
- Propietate pertsonalizatua **tempMedia**:
SELECT hiria, MAX(fecha) as fecha, AVG(temperatura) as temperatura, AVG(hezetasuna) as hezetasuna FROM FLOWFILE GROUP BY hiria

Edit Processor QueryRecord 2.11.0	
Settings	Properties
Required field +	
Property	Value
Record Reader	JsonTreeReader
Record Writer	ParquetRecordSetWriter
Include Zero Record FlowFiles	false
Default Decimal Precision	10
Default Decimal Scale	0
tempMedia	SELECT hiria, MAX(fecha) as fecha, A...

13. PufFile prozesadorea:

	PutFile Gold 10 Min
	PutFile 2.11.0
	org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

a. Propiedades atalean:

- **Directory: /opt/nifi/arietak/07-ariketa-aemet-data-lake/gold**
- **Conflict Resolution Strategy: replace**

Edit Processor PutFile 2.11.0	
Settings	Properties
Required field +	
Property	Value
Directory	/opt/nifi/arietak/07-ariketa-aemet-da...
Conflict Resolution Strategy	replace
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

Edit Processor PutFile 2.11.0	
Settings	Properties
Name*	PutFile Gold 10 Min
Id	bf983adc-01a0-1000-04d0-93434e2b64cd
Type	PutFile 2.11.0
Bundle	org.apache.nifi - nifi-standard-nar
Penalty Duration*	30 sec
Yield Duration*	1 sec
Bulletin Level*	WARN

14. PutMongoRecord prozesadorea:

	PutMongoRecord 7-kasua Gold PutMongoRecord 2.11.0 org.apache.nifi - nifi-mongodb-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

a. **Properties** atalean:

- **Client Service: MongoDBControllerService**
MongoDBControllerService konfiguratu eta abilitatu (*enable*):
 - **Connection String: mongodb://iabd-mongodb-nifi:27017**

Controller Service Details	
Settings	
Properties	
Required field	
Property	Value
Mongo URI	mongodb://iabd-mongodb-nifi:27017
Database User	No value set
Password	No value set
SSL Context Service	No value set
Write Concern	ACKNOWLEDGED

- **Mongo Database Name: iabd**
- **Mongo Collection Name: 7kasua-gold**
- **Record Reader: ParquetReader** (*enable*)

Edit Processor | PutMongoRecord 2.11.0

Settings
Scheduling
Properties

Required field

Property	Value
Client Service	MongoDBControllerService
Mongo Database Name	iabd
Mongo Collection Name	7kasua-gold
Record Reader	ParquetReader
Batch Size	100
Ordered	False
Bypass Validation	False
Update Key Fields	No value set

Edit Processor | PutMongoRecord 2.11.0

Settings
Scheduling
Properties

Name*
PutMongoRecord 7-kasua Gold

Id
bfa318d1-01a0-1000-98af-344c87c7e4ff

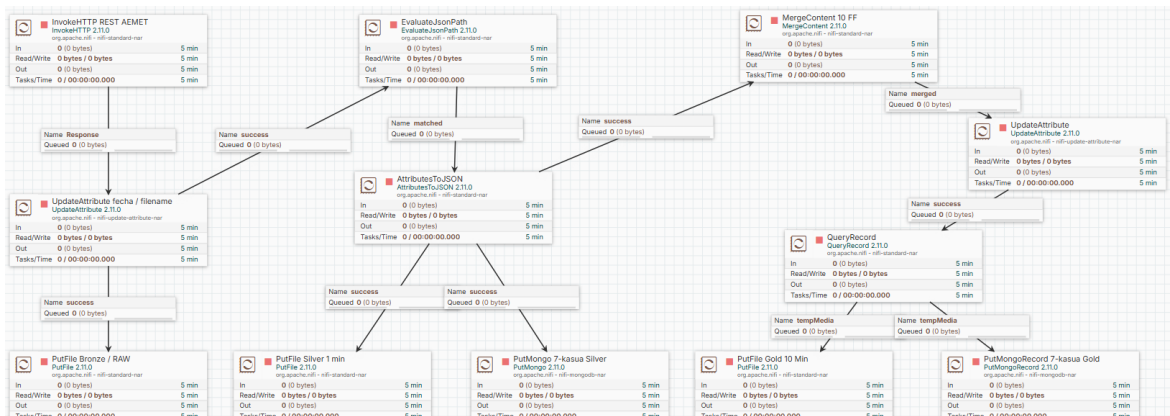
Type
PutMongoRecord 2.11.0

Bundle
org.apache.nifi - nifi-mongodb-nar

Penalty Duration 30 sec
Yield Duration 1 sec

Bulletin Level WARN

15. Konexioak ezarri:



JSON baten edukia ikusi:

docker exec iabd-nifi cat /opt/nifi/ariketak/07-ariketa-aemet-data-lake/silver/2026-09-20_17-44-15.json

```
PS C:\WINDOWS\System32> docker exec iabd-nifi cat /opt/nifi/ariketak/07-ariketa-aemet-data-lake/silver/2026-09-20_17-44-15.json
{"temperatura": "26", "fecha": "2026-09-20 17:44:15", "hiria": "Elche/Elx", "hezetasuna": "58"}
```

Hemen JSON garbitua ikusiko da.

Gold egiaztatu:

Gold karpetan json-ak gehitzen doazela ikusi:

docker exec iabd-nifi ls -lh /opt/nifi/ariketak/07-ariketa-aemet-data-lake/gold

```
PS C:\WINDOWS\System32> docker exec iabd-nifi ls -lh /opt/nifi/ariketak/07-ariketa-aemet-data-lake/gold
total 144K
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:24 2026-09-20_17-24-08.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:25 2026-09-20_17-25-08.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:26 2026-09-20_17-26-08.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:27 2026-09-20_17-27-09.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:28 2026-09-20_17-28-09.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:29 2026-09-20_17-29-09.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:30 2026-09-20_17-30-10.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:31 2026-09-20_17-31-10.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:32 2026-09-20_17-32-10.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:33 2026-09-20_17-33-10.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:34 2026-09-20_17-34-11.parquet
-rw-r--r-- 1 nifi nifi 1.4K Sep 20 17:35 2026-09-20_17-35-11.parquet
```

Kasu honetan parquet fitxategiak ikusiko dira (datu agregatuak).

Pasatu Windows-era parquet fitxategi bat, irakurri ahal izateko gero:

docker cp iabd-nifi:/opt/nifi/ariketak/07-ariketa-aemet-data-lake/gold/2026-09-20_17-44-16.parquet C:\nifi\ariketak\07-ariketa-aemet-data-lake\gold.parquet

```
PS C:\WINDOWS\System32> docker cp iabd-nifi:/opt/nifi/ariketak/07-ariketa-aemet-data-lake/gold/2026-09-20_17-44-16.parquet C:\nifi\ariketak\07-ariketa-aemet-data-lake\gold.parquet
Successfully copied 1.42kB (transferred 3.07kB) to C:\nifi\ariketak\07-ariketa-aemet-data-lake\gold.parquet
PS C:\WINDOWS\System32>
```

Parquet fitxategiak irakurri ahal izateko, Python-eko bi liburutegi instalatu:

pip install pandas pyarrow

```
PS C:\WINDOWS\System32> pip install pandas pyarrow
Collecting pandas
  Downloading pandas-3.0.6-cp314-cp314-win_amd64.whl.metadata (19 kB)
Collecting pyarrow
  Downloading pyarrow-25.0.1-cp314-cp314-win_amd64.whl.metadata (3.0 kB)
Collecting numpy>=2.3.3 (from pandas)
  Downloading numpy-2.5.3-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.6-cp314-cp314-win_amd64.whl (9.8 MB)
   9.8/9.8 MB 330.1 kB/s 0:00:29
Downloading pyarrow-25.0.1-cp314-cp314-win_amd64.whl (28.6 MB)
   28.6/28.6 MB 574.2 kB/s 0:00:56
Downloading numpy-2.5.3-cp314-cp314-win_amd64.whl (12.7 MB)
   12.7/12.7 MB 1.8 MB/s 0:00:06
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2026.4-py2.py3-none-any.whl (347 kB)
Installing collected packages: tzdata, six, pyarrow, numpy, python-dateutil, pandas
   3/6 [numpy] WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\ProgramData\Anaconda3\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed numpy-2.5.3 pandas-3.0.6 pyarrow-25.0.1 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.4
```

python -c "import pandas as pd; print(pd.read_parquet(r'C:\nifi\ariketak\07-ariketa-aemet-data-lake\gold.parquet').to_string(index=False))"

```
PS C:\WINDOWS\System32> python -c "import pandas as pd; print(pd.read_parquet(r'C:\nifi\ariketak\07-ariketa-aemet-data-lake\gold.parquet').to_string(index=False))"
      hiria      fecha  temperatura  hezetasuna
Elche/Elx  2026-09-20  17:44:15+00:00  26.000000000  58.000000000
```

- b. MongoDB-n egiaztatzeko, terminalean exekutatu:

docker exec -it iabd-mongodb-nifi mongosh

```
mongosh mongodh://127.0.0.1:27017
PS C:\WINDOWS\System32> docker exec -it iabd-mongodb-nifi mongosh
Current Mongosh Log ID: 6aae6b3939426df1f6258d6d
Connecting to: mongodh://127.0.0.1:27017?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.11.1
Using MongoDB: 8.3.11
Using Mongosh: 2.11.1
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-19T10:17:03.811+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-19T10:17:04.092+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-19T10:17:04.092+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-19T10:17:04.092+00:00: vm.max_map_count is too low
2026-09-19T10:17:04.092+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
```

Praktika honetako datu basean sartu:

use iabd

```
test> use iabd
switched to db iabd
```

show collections

```
iabd> show collections
6kasua
6kasua-record
7kasua-gold
7kasua-silver
```

Silver egiaztatu:

db["7kasua-silver"].find()

```
iabd> db["7kasua-silver"].find()
[
  {
    _id: ObjectId('6ab0199d9ea4c473f1ed4796'),
    temperatura: '26',
    fecha: '2026-09-20 17:36:29',
    hiria: 'Elche/Elx',
    hezetasuna: '58'
  },
  {
    _id: ObjectId('6ab0199d9ea4c473f1ed4797'),
    temperatura: '26',
    fecha: '2026-09-20 17:36:29',
    hiria: 'Elche/Elx',
    hezetasuna: '58'
  }
]
```

db["7kasua-silver"].countDocuments()

Gold egiaztatu:

db["7kasua-gold"].find()

```
iabd> db["7kasua-gold"].find()
[
  {
    _id: ObjectId('6ab0199d9ea4c473f1ed4797'),
    hiria: 'Elche/Elx',
    fecha: ISODate('2026-09-20T17:36:29.000Z'),
    temperatura: Decimal128('26.000000000'),
    hezetasuna: Decimal128('58.000000000')
  },
  {
    _id: ObjectId('6ab0199d9ea4c473f1ed4796'),
    hiria: 'Elche/Elx',
    fecha: ISODate('2026-09-20T17:36:29.000Z'),
    temperatura: Decimal128('26.000000000'),
    hezetasuna: Decimal128('58.000000000')
  }
]
```

db["7kasua-gold"].countDocuments()